

A REPORT

ON

DEEP LEARNING APPROACH FOR VIDEO VIOLENCE

DETECTION

BY

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ID No.: 2023AC05652

AT

Vasishta Nirman India Private Limited

Chennai, Tamil Nadu, India



BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI

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Artificial Intelligence & Machine
Learning

Prepared in partial fulfilment of the

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CERTIFICATE

This is to certify that the Project Work entitled DEEP LEARNING APPROACH FOR VIDEO VIOLENCE DETECTION and submitted by AKURATI KAUSTIKI ID No. 2023AC05652 in partial fulfillment of the requirements of AIML CZG628T Dissertation, embodies the work done by him/her under my supervision.

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Abstract:

This dissertation presents a deep learning approach for video violence detection in Indian CCTV surveillance environments. The system integrates violence classification, weapon detection, scene graph analysis, and action classification for comprehensive threat assessment.

For violence classification, three temporal architectures were evaluated: MobileNetV3-Large + BiLSTM + Attention, EfficientNet-B0 + BiLSTM + Attention, and 3D CNN. The best model achieved 97.7% accuracy on the RLVS benchmark (1,759 videos) with 99.9% AUC-ROC. Cross-domain evaluation revealed significant domain gap: RLVS-trained models achieved only 67% accuracy on Indian data, while domain-specific training on a curated Indian dataset (291 videos across 5 environmental contexts) achieved 80.4% test accuracy.

For weapon detection, approximately 676 images were annotated using Grounding DINO (Vision Transformer) for three classes: bladed weapons, firearms, and blunt weapons. Three architectures were compared: YOLOv11 (Small), Roboflow 3.0 (YOLOv8s), and Faster R-CNN (ResNet50+FPN). Roboflow 3.0 achieved the best performance with mAP@50 of 0.4978 and F1-score of 0.537, while one-stage detectors trained $8\times$ faster than two-stage Faster R-CNN. All custom-trained models achieved F1-scores of 0.50-0.54, representing a $6.7\times$ improvement over the best zero-shot model (YOLO-World-L at 0.08 F1), confirming that domain-specific training is essential for Indian weapon types.

Scene graph analysis extracts spatial relationships with proximity thresholds and optical flow-based motion features. Action classification combines X3D-M (Kinetics-400) with motion pattern analysis. This work demonstrates that domain-specific training is critical for deploying violence detection in Indian surveillance contexts.



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1. Introduction

1.1 Problem Statement

Violence in public spaces has increased significantly, making automated surveillance systems essential for real-time threat detection. Traditional CCTV monitoring relies on manual observation, which is prone to fatigue and delays. This dissertation develops an automated violence detection system using deep learning for video classification from surveillance footage.

The primary challenge lies in deploying systems trained on Western benchmark datasets to Indian surveillance environments, where different violence patterns, weapon types, and environmental conditions create significant distribution shift.

1.2 Challenges

Domain Gap: Models trained on Western datasets (RLVS, Hockey Fight) show significant performance degradation on Indian CCTV footage due to differences in violence patterns, clothing, environment, and camera quality.

Weapon Diversity: Indian contexts include traditional weapons like lathi (bamboo stick), sickle, and machete that are absent from standard detection datasets like COCO.

Real-time Requirements: Surveillance applications demand low-latency inference suitable for edge deployment.

Limited Labeled Data: Annotated Indian violence and weapon datasets are scarce, requiring efficient training strategies.

1.3 Objectives

1. Develop a high-accuracy violence classifier using temporal deep learning architectures.
2. Create a domain-specific weapon detector for Indian weapon types.
3. Quantify cross-domain performance gap between Western and Indian datasets.
4. Integrate scene graph analysis for spatial relationship modeling.
5. Implement action classification for specific violence type identification.

1.4 Contributions

This dissertation makes the following contributions:

1. Evaluation of 3 violence classification architectures achieving 97.7% accuracy on RLVS benchmark, surpassing prior state-of-the-art (91.03%).
2. Quantification of 30.7 percentage point domain gap (97.7% to 67%) when transferring from RLVS to Indian context.
3. Creation of 676-image Indian weapon dataset with 3 classes annotated using Grounding DINO.
4. Comprehensive comparison of three detection architectures (YOLOv11, Roboflow 3.0, Faster R-CNN) demonstrating one-stage detectors outperform two-stage by 18% mAP@50 while training $8\times$ faster, with $6.7\times$ improvement over zero-shot baselines.
5. Scene graph analysis for explainability, with empirical validation showing complementary (not discriminative) role to deep learning classification
6. Action classification combining X3D-M with motion pattern analysis.

2. Literature Review

2.1 Violence Detection Approaches

Recent research in violence detection has explored various deep learning approaches for surveillance applications. Merit et al. (2025) proposed a temporal fusion strategy combining MobileNetV3 with LSTM networks, achieving 91.03% accuracy on the RLVS dataset using slow fusion technique for better spatiotemporal feature extraction [1]. Senthilkumar et al. (2025) developed a hybrid system using YOLOv7 for human detection combined with MobileNet-BiLSTM for violence classification, achieving 91% accuracy with real-time alerting capability [2].

Jeevan et al. (2025) utilized YOLOv8's anchor-free detection architecture for violence detection, reporting 96% accuracy with high precision-recall balance [3]. Sethi et al. (2025) demonstrated YOLOv7's effectiveness for violence detection with 95.15% accuracy, outperforming Faster R-CNN and SSD models [4]. Chakraborty et al. (2024) compared multiple architectures including VGG16, VGG19, MobileNetV2, YOLOv8, and YOLO-NAS, finding that YOLOv8 performed best with 0.884 mAP50 for violence detection in surveillance footage [5].

Varghese et al. (2025) proposed a temporal-spatial deep learning framework leveraging dynamic 3D attention maps, demonstrating the importance of attention mechanisms in capturing discriminative spatiotemporal features for violence detection [6]. Park et al. (2024) developed a Conv3D-based violence detection network using both optical flow and RGB data, showing that multi-stream approaches can enhance detection performance [7].

These studies highlight the effectiveness of combining CNN-based spatial feature extractors with temporal modeling (LSTM/BiLSTM) for video-based violence detection, which forms the foundation of the approach adopted in this dissertation.

2.2 Research Gaps

The literature review reveals several critical gaps that this dissertation addresses:

1. **Domain Adaptation Gap:** Existing violence detection systems achieve high accuracies on Western benchmark datasets (RLVS, Hockey Fight, Movies) but are not validated for Indian surveillance contexts with different environmental settings, clothing patterns, and violence characteristics.
2. **Weapon Detection Integration:** Current approaches lack weapon detection integration, which could enable early threat identification before violence occurs. No existing work combines violence classification with domain-specific weapon detection for Indian contexts.
3. **Cross-Domain Evaluation:** No existing work quantifies the performance degradation when models trained on Western datasets are deployed in Indian surveillance environments, leaving the domain gap uncharacterized.

2.3 Object Detection for Weapons

Existing weapon detection research focuses primarily on firearms and knives common in Western contexts. The COCO dataset includes only knife and scissors categories. No comprehensive dataset covers Indian traditional weapons like lathi, sickle, or machete.

Zero-shot detection models (YOLO-World, Grounding DINO, OWL-ViT) offer text-prompted detection but struggle with domain-specific weapons not well-represented in training data.

3. System Architecture

3.1 Overall Pipeline

The system uses a shared base sampling window of 32 sampled frames ($\text{frame_skip} = 2$). Components operate on per-component strides: weapon detection runs on every 2nd sampled frame, action classification runs on every sampled frame but uses a 16-frame clip, and scene graph analysis runs on every 4th sampled frame. Action classification is triggered only when the violence classifier detects violence above the threshold.

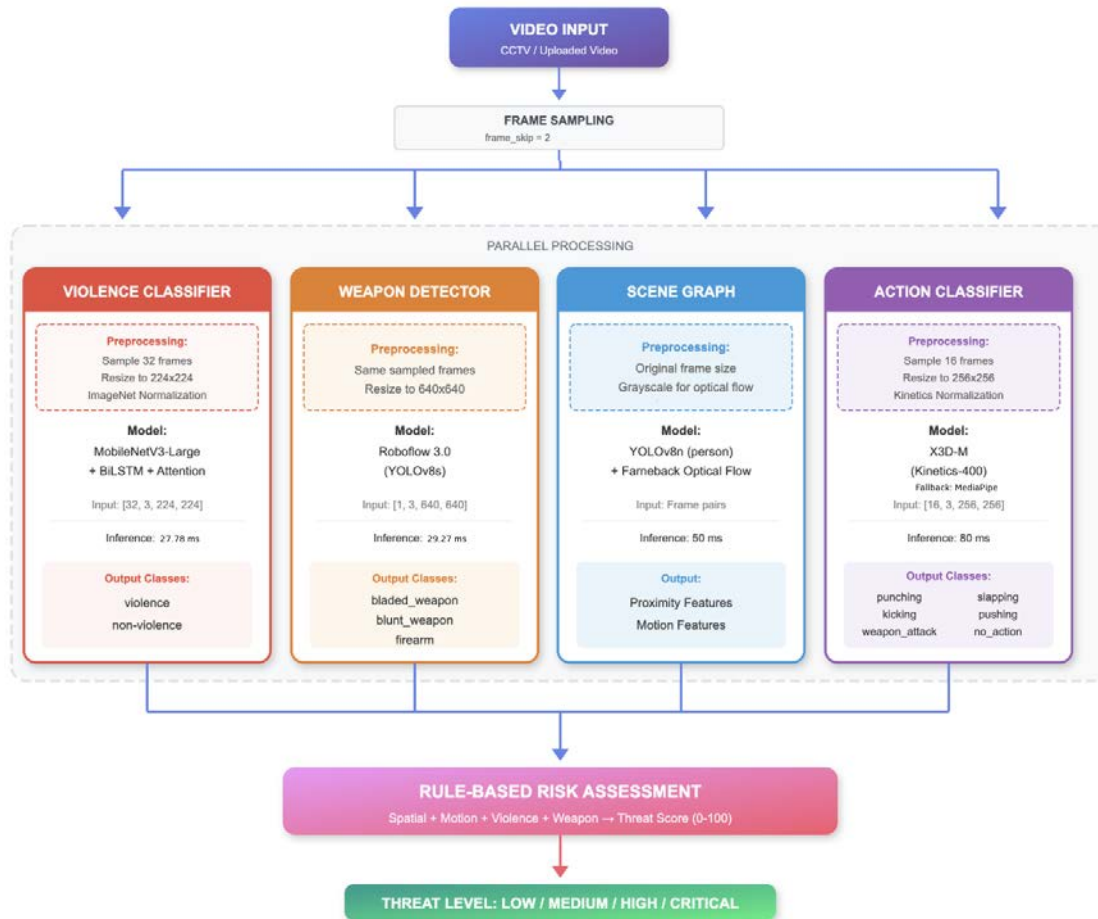


Figure 3.1: Overall system architecture

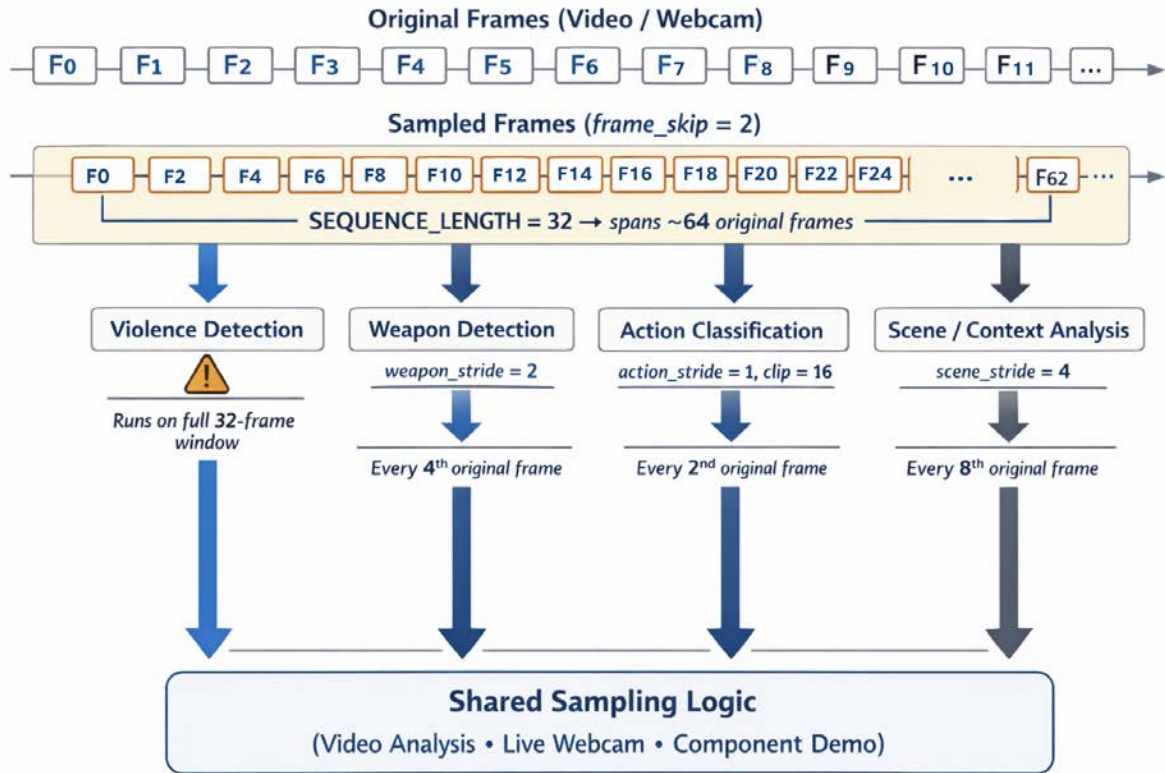


Figure 3.2: Frame Input

3.2 Component Specifications

Table 3.1: System Component Specifications

Component	Model	Input	Inference Time
Violence Classifier	MobileNetV3-BiLSTM-Attention	32 sampled frames @ 224x224 (frame_skip = 2)	27.78 ms
Weapon Detector	Roboflow 3.0 (YOLOv8s)	Single frame @ 640x640 (stride = 2 over sampled frames)	29.27 ms
Scene Graph	YOLOv8n + Optical Flow	Single frame (optical flow between consecutive sampled frames; stride = 4)	~50 ms
Action Classifier	X3D-M + Motion Analysis	16 sampled frames @ 256x256 (clip_len = 16, stride = 1)	~80 ms

3.3 Risk Assessment Module

The risk assessment module synthesizes outputs from all parallel processing components to generate an interpretable threat score ranging from 0 to 100. Rather than employing a learned fusion approach, the system uses a rule-based scoring mechanism that aggregates evidence from spatial analysis, motion dynamics, and violence classification confidence. Figure 3.2 presents the complete scoring logic and threshold definitions.

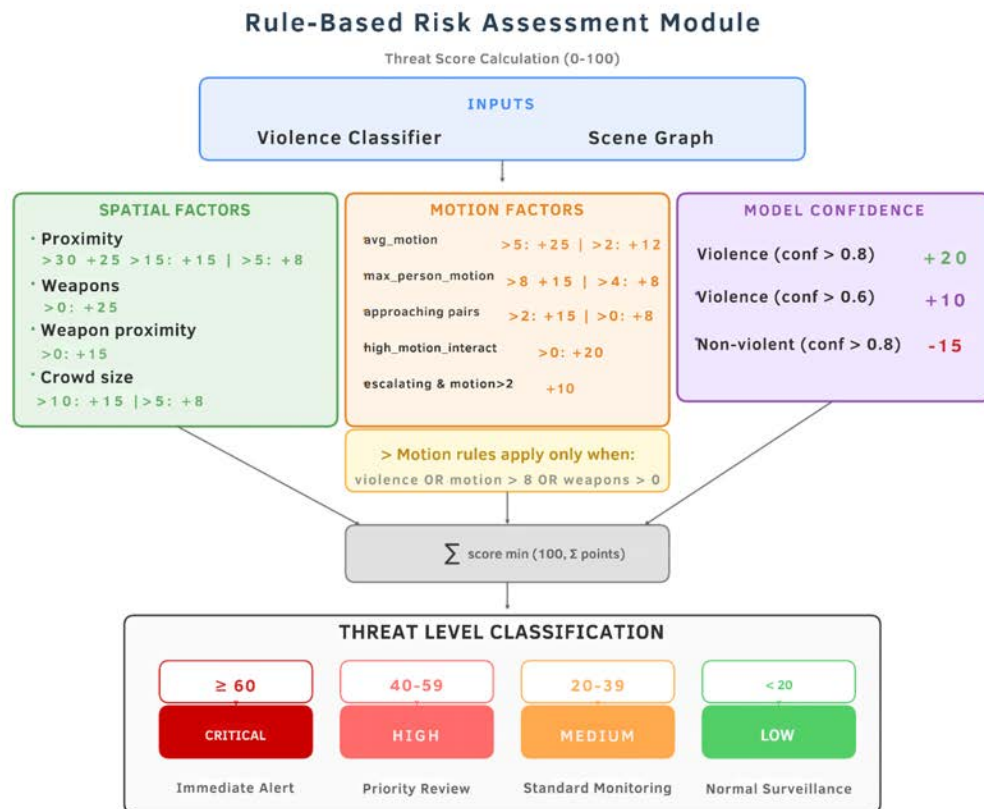


Figure 3.3: Risk Assessment Diagram

3.3.1 Scoring Mechanism

The threat score accumulates points from three categories of factors:

Spatial factors assess the physical configuration of the scene, including interpersonal proximity (measured as close interactions from the scene graph), crowd density, weapon presence, and person-weapon proximity. These factors capture static risk indicators that persist across frames.

Motion factors evaluate dynamic behavior through optical flow analysis, including average motion intensity, maximum individual motion, approaching trajectories, and high-motion interactions indicative of physical altercations. Critically, motion factors are conditionally applied only when contextual evidence suggests potential violence, specifically, when the violence classifier detects violence, when average motion exceeds 8.0 (indicating unusually high activity), or when weapons are detected. This conditional gating prevents benign high-motion scenarios (such as sports or celebrations) from triggering false alarms.

Model confidence adjustments allow the violence classification model to modulate the final score. High-confidence violence predictions (+20 points for confidence > 0.8) reinforce threat assessment, while high-confidence non-violent predictions (-15 points) suppress scores that may have accumulated from ambiguous spatial or motion patterns.

3.3.2 Threshold Definitions

Proximity thresholds use normalized bounding box distances: very close (<0.15) indicates potential contact range, close (<0.25) represents arm's reach, near (<0.40) denotes nearby proximity, and far (≥ 0.40) indicates distant individuals. Motion thresholds are based on optical flow magnitude: values exceeding 5.0 indicate aggressive or violent movement, values above 2.0 suggest elevated activity, and values below 0.5 represent stationary subjects.

3.3.3 Threat Level Classification

The accumulated score maps to four discrete threat levels: CRITICAL (≥ 60), HIGH (40–59), MEDIUM (20–39), and LOW (< 20). Each level corresponds to an operational response appropriate for security monitoring contexts, ranging from immediate alerts requiring human intervention to standard passive surveillance.

3.4 Design Rationale

The rule-based approach was selected over learned fusion methods for several practical and methodological reasons:

Interpretability: Security operators must understand why the system generates alerts. Explicit scoring rules with named factors (e.g., "proximity > 30 : +25 points") enable operators to verify system reasoning and make informed decisions about intervention. This transparency is essential for deployment in safety-critical surveillance contexts where blind trust in opaque models is inappropriate.

Adjustability: Operational requirements vary across deployment environments—a hospital may require different sensitivity thresholds than a public transit station. Rule-based scoring allows security administrators to tune individual thresholds or point values without retraining neural network components, enabling rapid adaptation to site-specific needs.

Robustness to limited data: Training a learned fusion model would require substantial multi-modal annotated data combining violence labels, weapon annotations, scene graphs, and threat level ground truth. Such comprehensive datasets are scarce, and the custom Indian dataset (300 videos) is insufficient for learning robust fusion weights. The rule-based approach sidesteps this limitation by encoding domain knowledge directly.

Modularity and failure isolation: Each component—violence classifier, weapon detector, scene graph analyzer—operates independently and contributes additively to the threat score. This architecture allows individual components to be updated, replaced, or disabled without affecting others, and ensures that failure in one module degrades performance gracefully rather than catastrophically.

Conditional context-awareness: The gating mechanism for motion factors represents a key design decision. Early experiments revealed high false positive rates in scenarios with legitimate high-motion activity. By conditioning motion scoring on violence detection or weapon presence, the system maintains sensitivity to genuine threats while suppressing spurious alerts from benign activity

3.4.1 Threshold Calibration

Empirical analysis on the Indian dataset (50 videos per class) confirmed that proximity and motion features do not independently discriminate between violence and non-violence classes ($p > 0.05$, Youden's $J \approx 0$). This validates the architectural decision to treat scene graph analysis as a supplementary contextual signal rather than a primary classifier. Thresholds are therefore physically grounded: VERY_CLOSE (0.15) corresponds to contact range, CLOSE (0.25) to arm's reach, and HIGH_MOTION (5.0) to aggressive movement patterns observable in optical flow magnitude

4. Violence Classification Module

4.1 Training Pipeline

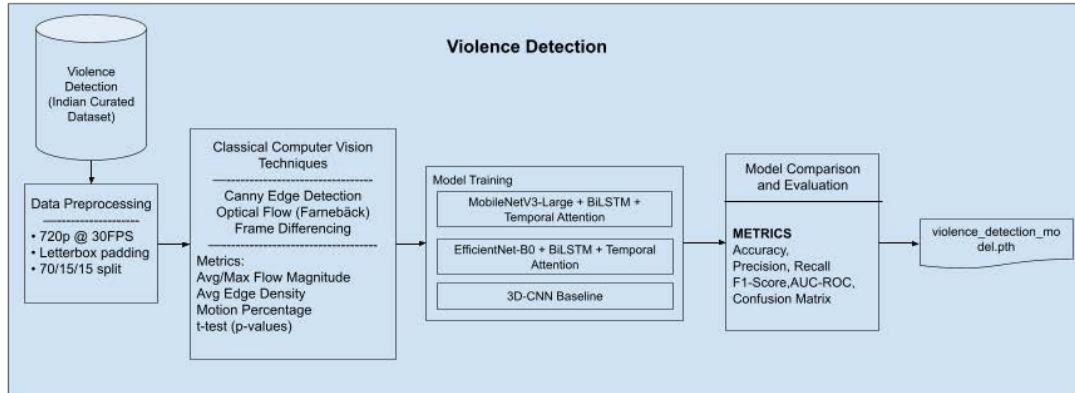


Figure 4.1: Violence Detection Training pipeline

4.2 Dataset Description

4.2.1 RLVS Dataset (Benchmark Dataset)

Table 4.1: RLVS Dataset Properties

Property	Details
Total Videos	1,759
Classes	Violent (1,000) / Non-Violent (759)
Video Duration	3-10 seconds
Processed Resolution	720p @ 30 FPS
Split	Train (80%) / Val (10%) / Test (10%)

4.2.2 Indian Context Dataset (Primary Dataset)

A custom dataset of 150 violent videos was curated from Indian sources to capture region-specific violence patterns and surveillance characteristics.

Overview:

Table 4.2: Indian Dataset Properties

Property	Details
Total Videos	150 (Violent Class)
Video Duration	6-11 seconds
Sources	YouTube (90.7%), Indian Movies (6%), Instagram (1.3%), News (1.3%)
Original Resolution	Variable (608×1080, 720×1280, 1280×720, 1920×1080)
Original Frame Rate	Variable (24-30 FPS)
Codec	H.264 (MP4)
Purpose	Training violence classification models for Indian surveillance scenarios

Environment Types:

The dataset covers five distinct surveillance environments commonly found in Indian settings:

Table 4.3: Indian Dataset Environment Distribution

Environment	Videos	Percentage	Description
Residential Area	32	21.3%	Apartments, housing colonies, neighborhoods
Road	31	20.7%	Streets, highways, footpaths
Public Transport	31	20.7%	Buses, trains, metros, railway stations
Market/Shops	29	19.3%	Retail stores, shopping areas, bazaars
Educational Institutes	26	17.3%	Schools, colleges, campus areas

Dataset Characteristics:**Table 4.4: Indian Dataset Characteristics**

Factor	Description
Geographic Context	Indian streets, markets, transport systems
Camera Quality	Mixed (CCTV-style SD to HD smartphone footage)
Crowd Density	Variable (sparse to highly crowded Indian settings)
Clothing/Appearance	Indian attire, diverse demographics
Lighting Conditions	Daylight, low-light, indoor/outdoor

4.2.3 Preprocessing Pipeline

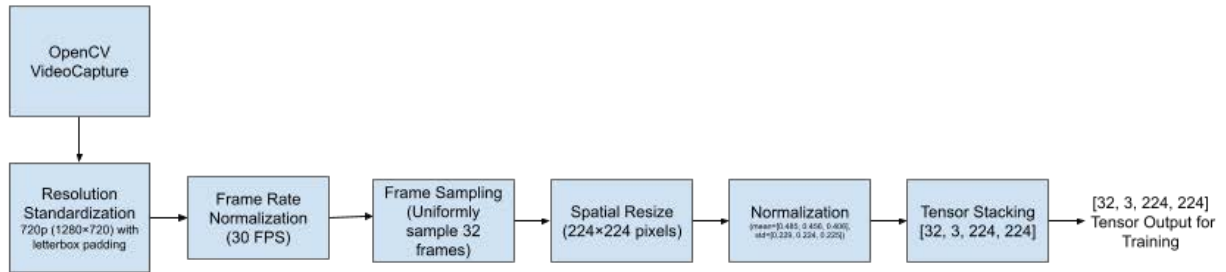


Figure 4.2: Preprocessing Pipeline

Table 4.5: Video Preprocessing Pipeline

Step	Operation	Details
1	Resolution Standardization	Resize to 720p (1280×720) with letterbox padding
2	Frame Rate Normalization	Resample to 30 FPS
3	Codec Conversion	Re-encode using MJPG codec
4	Frame Sampling	Uniformly sample 32 frames (frame_skip=2 → 15 FPS effective)
5	Spatial Resize	Resize frames to 224×224 pixels
6	Normalization	ImageNet stats (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225])
7	Tensor Stacking	Stack into tensor [32, 3, 224, 224]

4.3 Classical Computer Vision Analysis (Exploratory)

Before training deep learning models, classical CV techniques were applied to understand motion patterns in violent vs non-violent scenes.

Table 4.6: Classical CV Techniques Overview

Technique	Purpose	Output
Canny Edge Detection	Detect object/body contours (hands, faces, weapons)	Binary edge map (Avg Edge Density)
Optical Flow (Farneback)	Capture motion intensity and direction between frames	Flow magnitude + angle
Frame Differencing	Detect sudden motion between consecutive frames	Binary motion mask

Table 4.7: Classical CV Analysis Results on RLVS Dataset

Metric	Violent (Mean $\pm$ SD)	Non-Violent (Mean $\pm$ SD)	t-statistic	p-value	Significant
Avg Flow Magnitude	2.0094 $\pm$ 1.6603	1.0794 $\pm$ 1.4854	1.86 $\times$	0.069654	No
Max Flow Magnitude	6.3358 $\pm$ 4.5913	3.3877 $\pm$ 2.8640	2.44 $\times$	0.019629	Yes
Avg Edge Density	0.0144 $\pm$ 0.0104	0.0172 $\pm$ 0.0080	-0.94 $\times$	0.356410	No

Table 4.8: Classical CV Analysis Results on Indian Dataset

Metric	Violent (Mean $\pm$ SD)	Non-Violent (Mean $\pm$ SD)	t-statistic	p-value	Significant
Avg Flow Magnitude	2.72 $\pm$ 2.35	1.36 $\pm$ 1.42	2.219	0.0325	Yes
Max Flow Magnitude	6.96 $\pm$ 5.16	5.15 $\pm$ 3.54	1.289	0.2051	No
Avg Edge Density	0.020 $\pm$ 0.012	0.036 $\pm$ 0.028	-2.363	0.0234	Yes (Reversed)

4.4 Deep Learning Architecture Design

4.4.1 Models Compared

Table 4.9: Violence Classification Model Architectures

Model	Backbone	Temporal Module	Parameters	Target
Model 1	MobileNetV3-Large	BiLSTM + Attention	7.2M	Edge deployment
Model 2	EfficientNet-B0	BiLSTM + Attention	8.9M	Server-side
Model 3	3D CNN (C3D-style)	3D Convolutions	4.0M	Baseline

4.4.4 Model Selection Justification

Table 4.10: Model Selection Justification

Model	Justification
Model 1: MobileNetV3-Large + BiLSTM	MobileNetV3-Large is optimized for mobile and edge devices using depthwise separable convolutions and squeeze-and-excitation blocks, providing excellent accuracy-to-efficiency ratio. Combined with BiLSTM for temporal modeling and attention for frame importance weighting, this architecture is ideal for real-time CCTV deployment on resource-constrained edge devices.
Model 2: EfficientNet-B0 + BiLSTM	EfficientNet-B0 uses compound scaling to balance network depth, width, and resolution, achieving superior feature extraction compared to MobileNet. The higher parameter count (8.9M vs 7.2M) provides stronger spatial representations, making it suitable for server-side deployment where computational resources are less constrained and accuracy is prioritized.
Model 3: 3D CNN (C3D-style)	3D CNN serves as a baseline that processes spatiotemporal features jointly through 3D convolutions, representing the traditional approach to video classification. Unlike Models 1 and 2 which use ImageNet pretrained weights, 3D CNN is trained from scratch, enabling fair comparison to demonstrate the value of transfer learning on limited video datasets.

4.5 Training Configuration

Table 4.11: Training Hyperparameters

Parameter	Model 1 & 2	Model 3
Input Shape	[B, 32, 3, 224, 224]	[B, 3, 32, 224, 224]
Batch Size	4	4
Epochs	30	30
Optimizer	Adam (lr=1e-4)	Adam (lr=5e-4)
Weight Decay	1e-5	1e-5
Scheduler	ReduceLROnPlateau (factor=0.5, patience=3)	ReduceLROnPlateau (factor=0.5, patience=3)
Early Stopping	Patience: 7 epochs	Patience: 7 epochs
Loss Function	CrossEntropyLoss	CrossEntropyLoss
GPU	NVIDIA Tesla T4 (15.8 GB)	NVIDIA Tesla T4 (15.8 GB)

The input shape of **32 uniformly sampled frames at 224×224 resolution** was chosen to capture sufficient temporal information for violence detection, while matching the standard input size of ImageNet-pretrained backbones. A **batch size of 4** was selected to maximize GPU utilization within the Tesla T4’s 15.8 GB memory limit, since video data is memory-intensive (approximately 4.6 MB per sample).

The **Adam optimizer** was used for its adaptive learning rate capability. A **learning rate of 1e-4** was applied for pretrained models (MobileNetV3, EfficientNet) to avoid disrupting learned features, while a **higher learning rate of 5e-4** was used for the 3D CNN trained from scratch to

allow faster convergence. **Weight decay (1e-5)** was included for mild regularization to reduce overfitting.

A **ReduceLROnPlateau scheduler** was employed to halve the learning rate if validation loss did not improve for 3 consecutive epochs, enabling better convergence. **Early stopping** with a patience of 7 epochs was used to terminate training when validation performance stopped improving, preventing overfitting.

Finally, **Cross-Entropy Loss** was used as the classification objective, as the dataset had balanced class distribution.

4.6 Results

4.6.1 RLVS Dataset Results

Table 4.12: Violence Classification Results on RLVS Test Set

Model	Architecture	Parameters	Train Acc (%)	Test Acc (%)	Precision (%)	Recall (%)	F1-Score (%)	AUC - ROC (%)	Inference Time (ms)	Epochs
Model 1	MobileNetV3-Large + BiLSTM + Attention	7,241,011	99.57	97.73	97.73	97.73	97.73	99.91	23.08	13
Model 2	EfficientNet-B0 + BiLSTM + Attention	8,931,967	99.72	96.59	96.59	96.59	96.59	99.38	39.08	19
Model 3	3D CNN (C3D-style)	11,864,194	90.76	89.20	89.25	89.20	89.15	96.59	80.54	23

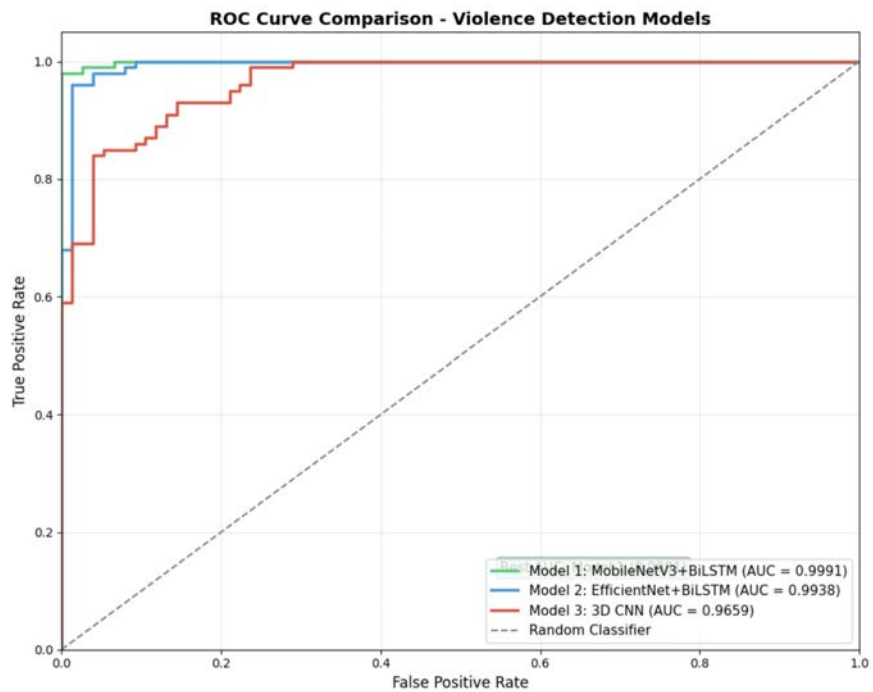
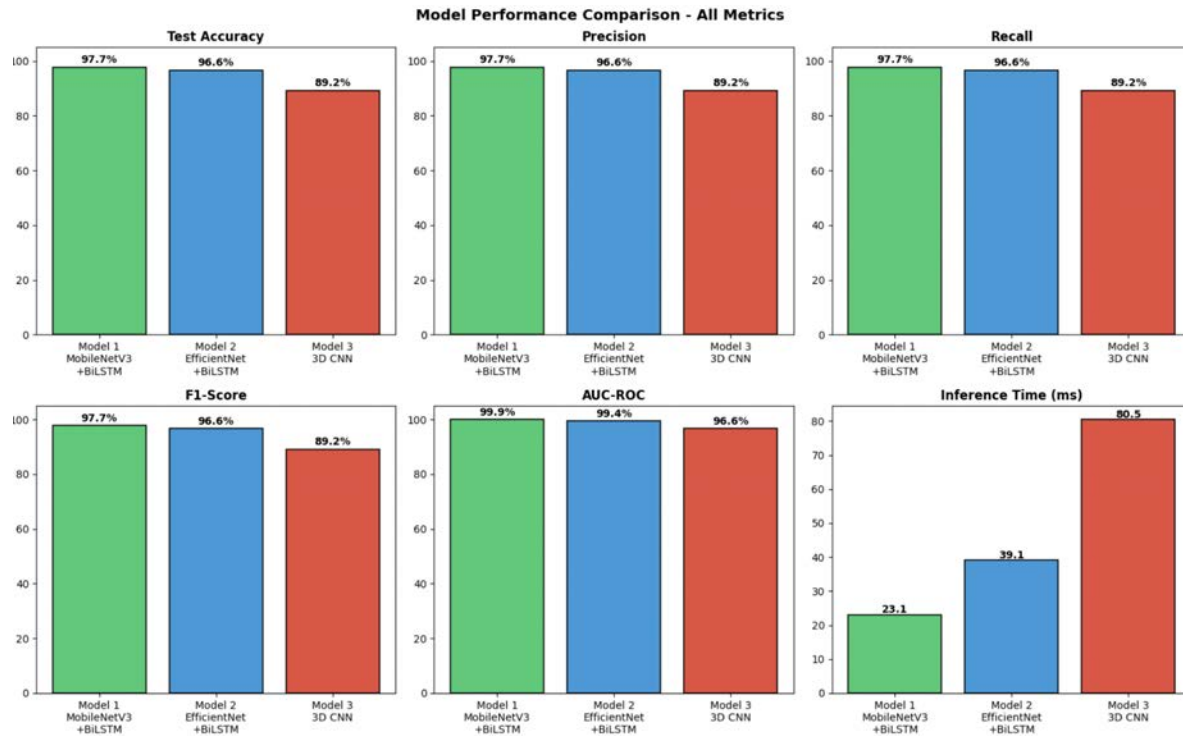


Figure 4.3: Model Performance Comparison on RLVS Dataset

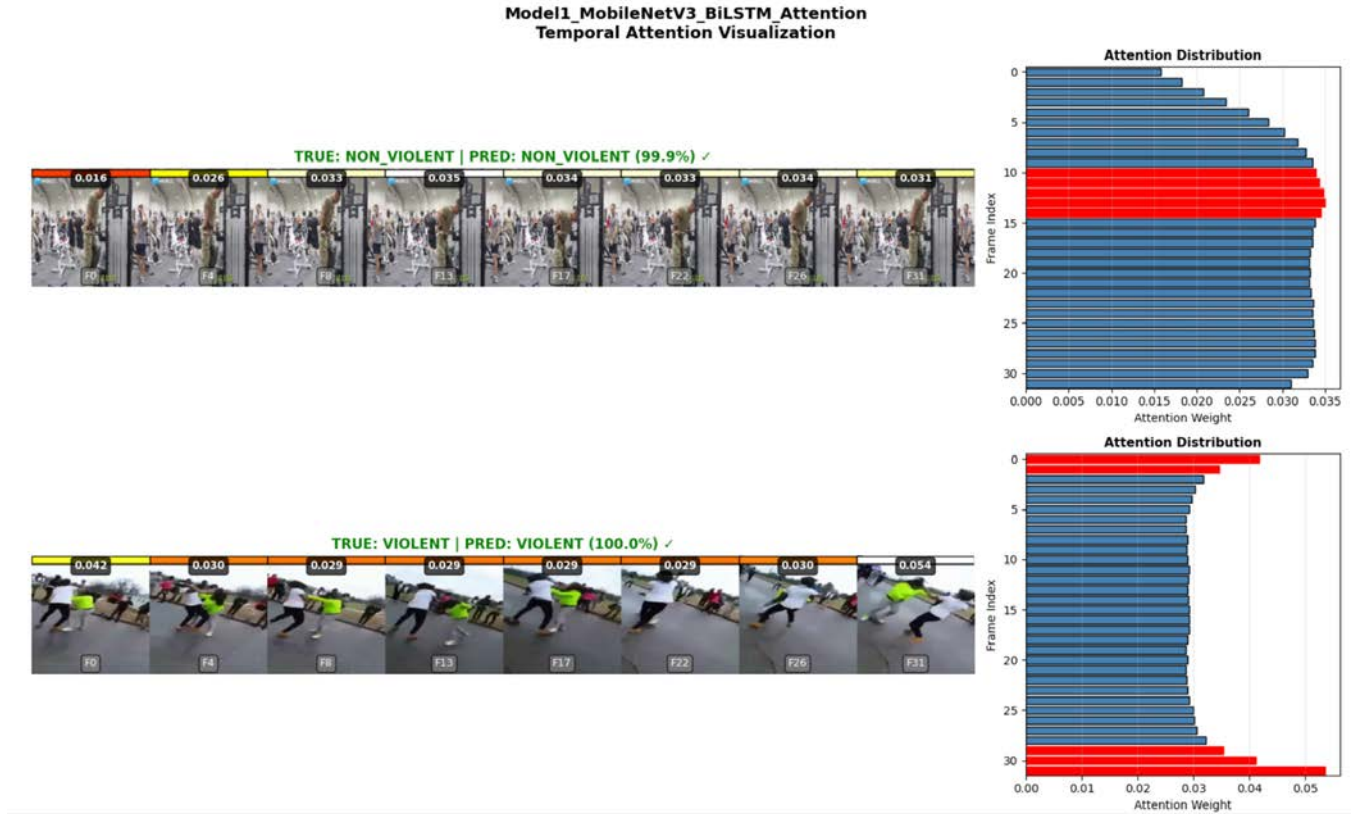


Figure 4.4: Attention Map Example on sample frames from RLVS Dataset

4.6.2 Indian Dataset Results

Table 4.13: Violence Classification Results on Indian Test Set

Model	Architecture	Parameters	Train Acc (%)	Test Acc (%)	Precision (%)	Recall (%)	F1-Score (%)	AUC-ROC (%)	Inference Time (ms)	Epochs
Model 1	MobileNetV3-Large + BiLSTM + Attention	7,241,011	96.53	80.43	81.94	80.43	80.20	85.63	27.78	16
Model 2	EfficientNet-B0 + BiLSTM + Attention	8,931,967	96.53	63.04	63.27	63.04	62.89	71.27	45.62	9

Model 3	3D CNN (C3D-style)	11,864,194	68.32	67.39	68.25	67.39	67.00	68.81	89.44	23
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Note: Indian dataset split - Train: 202, Val: 43, Test: 46 videos

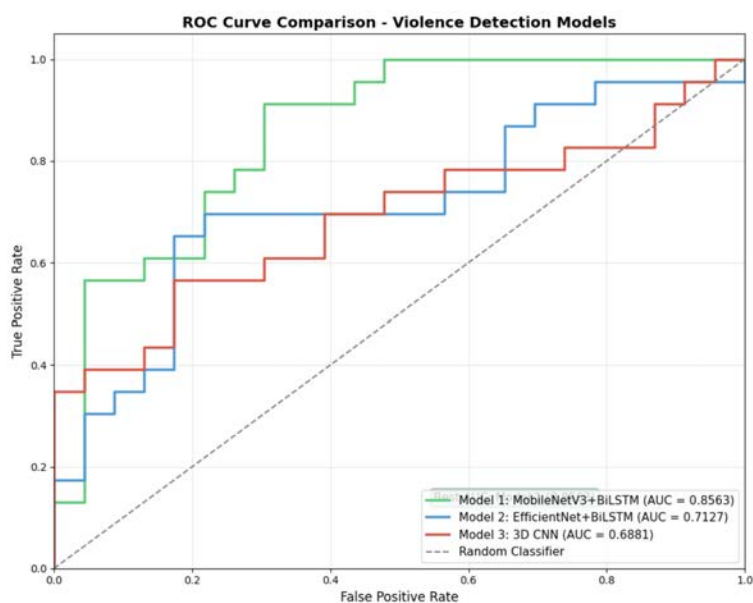
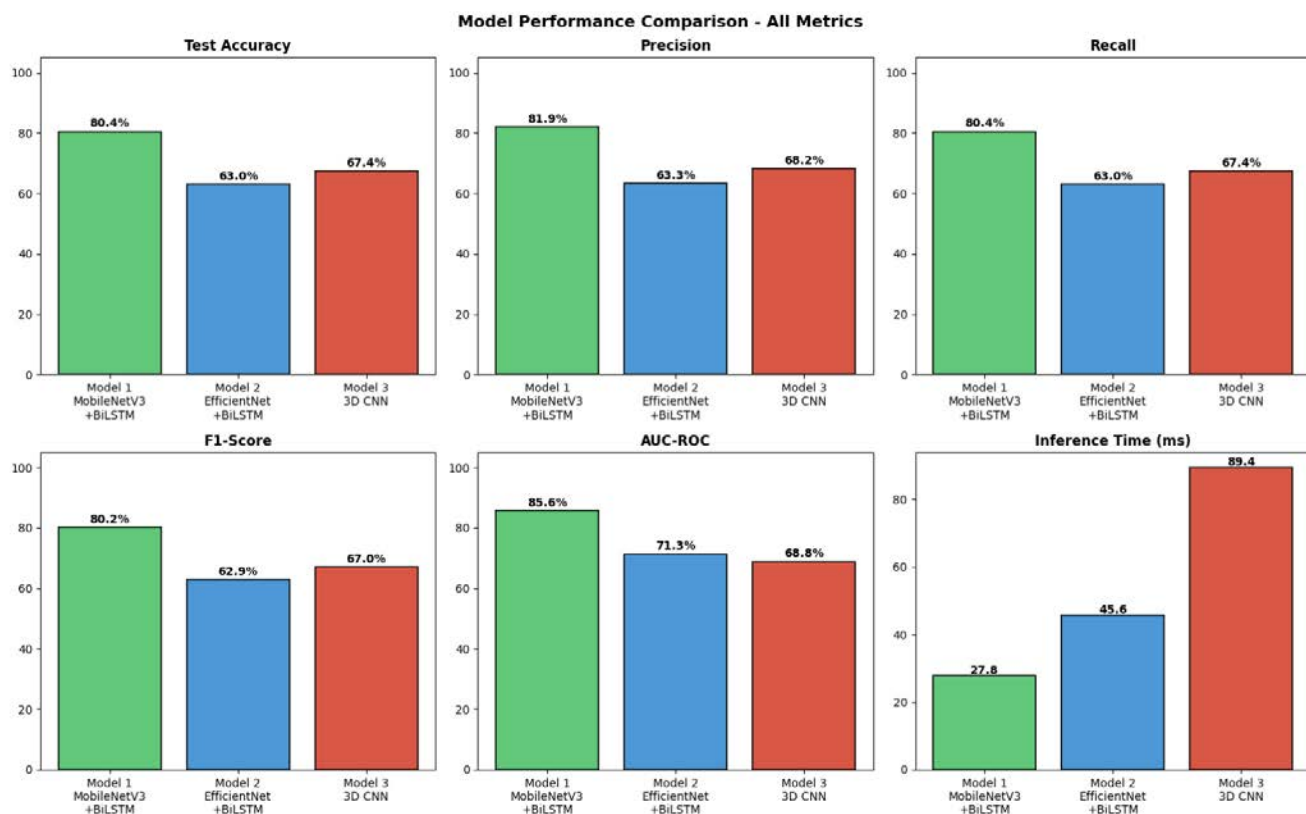


Figure 4.9: Model Performance Comparison on Indian curated Dataset

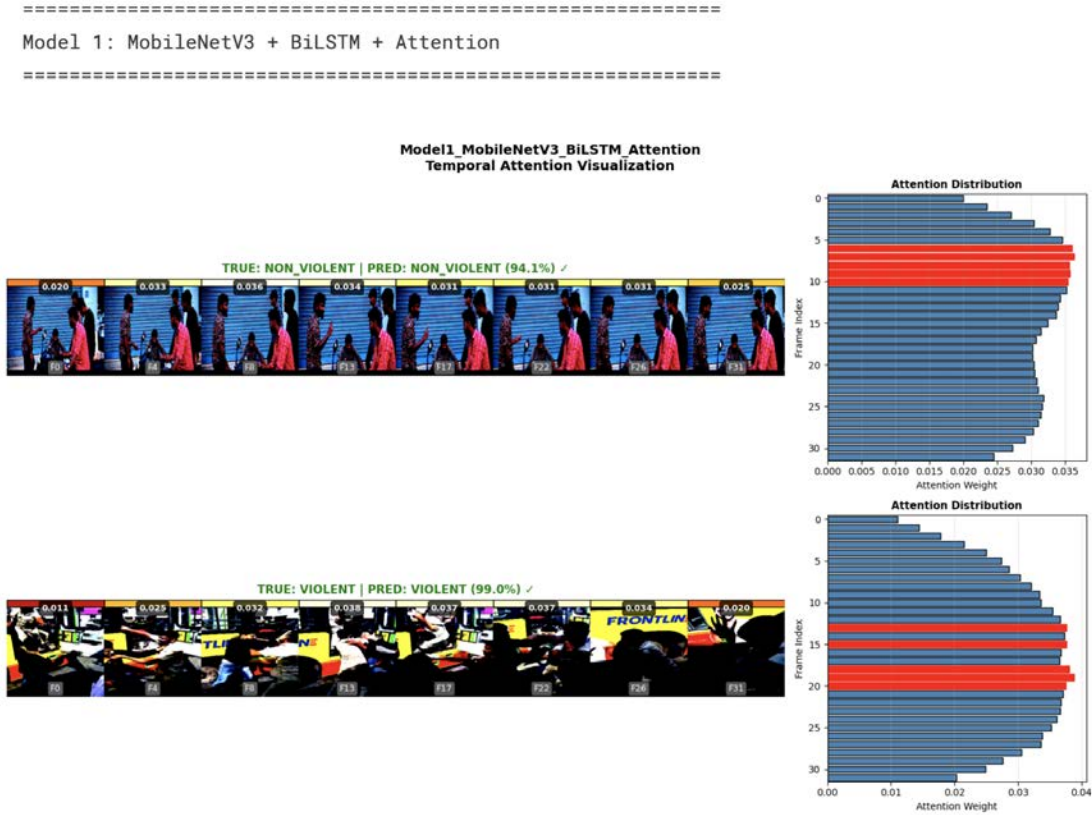


Figure 4.5: Attention Map Example on sample frames from RLVS Dataset

4.7 Key Findings

1. **Transfer learning outperforms training from scratch:** Models 1 & 2 (ImageNet pretrained) outperform Model 3 (trained from scratch) by 8.5 percentage points on RLVS dataset.
2. **State-of-the-art performance:** Best accuracy of 97.73% on RLVS significantly exceeds prior state-of-the-art: Merit et al. (91.03%) [1] and Senthilkumar et al. (91%) [2].
3. **Speed-accuracy trade-off:** MobileNetV3 backbone achieves best accuracy with fastest inference (23.08- 27.78 ms), making it suitable for real-time deployment.

4. **Significant discrimination:** All transfer learning models achieve >99% AUC-ROC on RLVS, indicating good class separation.
5. **Domain gap exists:** Performance drops from 97.73% (RLVS) to 80.43% (Indian) when training on domain-specific data, highlighting the importance of representative training data.

5. Weapon Detection Module

5.1 Indian Weapon Dataset

A custom dataset was created to address the lack of Indian weapon imagery in existing detection benchmarks. 676 Images were collected from CCTV footage, news sources, and film scenes.

Table 5.1: Weapon Dataset Statistics

Split	Images	Annotations
Train	541	816
Validation	68	116
Test	67	90

5.1.1 Class Definitions

Table 5.2: Weapon Class Taxonomy

Class ID	Class Name	Items Included	Annotations
0	bladed_weapon	knife, machete, sword, sickle, axe, dagger, switchblade, cleaver	372
1	blunt_weapon	lathi, iron rod, bat, cricket bat, hammer, crowbar, chain, brick	283
2	firearm	pistol, revolver, rifle, shotgun, automatic rifle	367



Figure 5.1: Sample annotations from Indian weapon dataset

5.2 Annotation Pipeline

Grounding DINO (Vision Transformer) was used for semi-automatic annotation with 68 text prompts covering weapon variations. Annotations were manually verified and corrected, then exported to YOLO format.



Figure 5.2: Semi-automatic annotation pipeline using Grounding DINO

5.3 Weapon Detection Model Training and Comparison

5.3.1 Models Compared

Three state-of-the-art object detection architectures were evaluated for weapon detection on the Indian dataset:

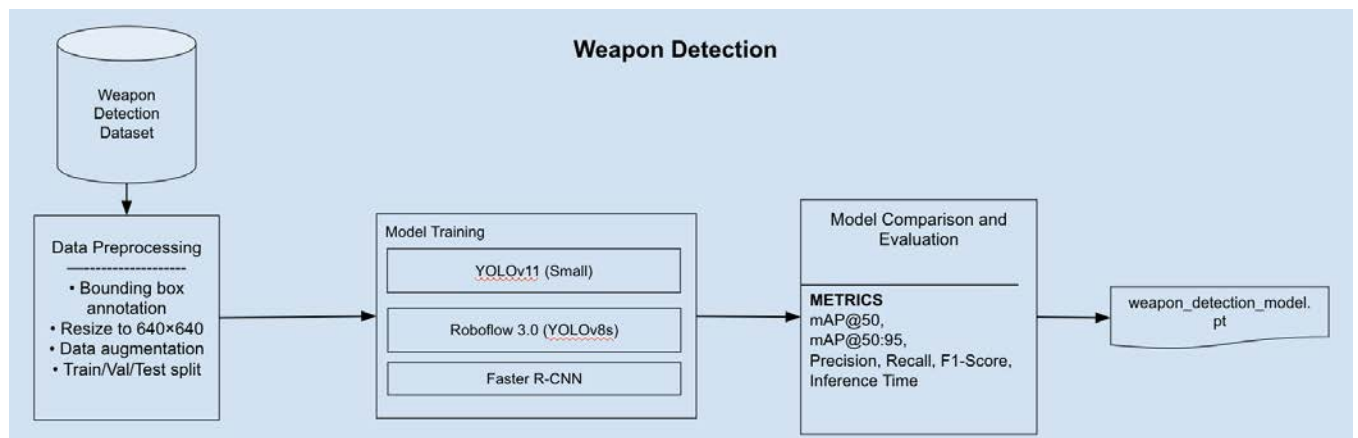


Figure 5.3: Weapon Detection pipeline

Table 5.3: Weapon Detection Model Architectures

Model	Type	Architecture	Parameters
YOLOv11 (Small)	One-stage	CSPDarknet + PANet	~9.4M
Roboflow 3.0 (YOLOv8s)	One-stage	CSPDarknet + PANet + Enhancements	~11.2M
Faster R-CNN	Two-stage	ResNet50 + FPN	~41.3M

Model Selection Justification:

YOLOv11 (Small): Latest release from Ultralytics (2024), successor to YOLOv8 with improved architecture and faster inference. Selected to evaluate cutting-edge one-stage detection performance.

Roboflow 3.0 (YOLOv8s): YOLOv8-compatible architecture with Roboflow's custom performance enhancements including improved augmentation strategies. Recommended by Roboflow for production deployments.

Faster R-CNN (ResNet50+FPN): Two-stage detector baseline using Region Proposal Network with Feature Pyramid Network backbone. Included to compare one-stage vs two-stage detection paradigms.

5.3.2 Training Configuration**Table 5.4: Weapon Detection Training Configuration**

Parameter	YOLOv11	Roboflow 3.0 (Yolov8s)	Faster R-CNN
Input Size	640 × 640	640 × 640	640 × 640

Batch Size	16	16	4
Epochs	50	50	30
Optimizer	MuSGD (lr=0.00143, momentum=0.9)	MuSGD (lr=0.00143, momentum=0.9)	SGD (lr=0.005, momentum=0.9)
Weight Decay	0.0005	0.0005	0.0005
Scheduler	Linear Decay	Linear Decay	StepLR (step=10, gamma=0.1)
Augmentation	Mosaic	Mosaic, MixUp, Copy-Paste	None
Pretrained Weights	COCO	COCO	COCO

Input Size and Batch Setup:

An input resolution of 640×640 was chosen to balance detection accuracy and computational cost, providing sufficient spatial detail to identify weapons in surveillance footage. Batch sizes were set based on the Tesla T4 (16 GB) GPU memory limit. YOLO-based models used a batch size of 16 due to their memory-efficient single-stage design, while Faster R-CNN used a smaller batch size of 4 because its two-stage architecture with FPN backbone requires higher memory per image.

Optimizer and Learning Rate:

For both YOLO variants, the Ultralytics framework automatically selected the MuSGD optimizer with a learning rate of 0.00143 and momentum 0.9, using an internally computed formula optimized for small datasets. Faster R-CNN was trained using SGD with learning rate 0.005 and momentum 0.9, following standard PyTorch detection settings. A StepLR scheduler reduced the learning rate by a factor of 0.1 every 10 epochs to support stable fine-tuning.

Regularization and Training Duration:

A weight decay of 0.0005 was applied to all models to reduce overfitting, which is especially important given the limited 541-image training set. The YOLO models were trained for 50 epochs, while Faster R-CNN was trained for 30 epochs, as its per-epoch training time was substantially higher.

Data Augmentation and Transfer Learning:

All models were initialized with COCO-pretrained weights, enabling effective transfer learning with limited data. Roboflow 3.0 (YOLOv8s) used strong augmentations (Mosaic, MixUp, Copy-Paste) to increase training diversity. YOLOv11 used Mosaic augmentation with default settings, while Faster R-CNN was trained without augmentation to serve as a baseline for architectural comparison.

5.3.3 Results

Table 5.5: Weapon Detection Results on Indian Test Set

Model	Training Time (min)	mAP@50	mAP@50-95	Precision	Recall	F1-Score
YOLOv11 (Small)	10.76	0.4625	0.2712	0.6079	0.4824	0.5379
Roboflow 3.0 (YOLOv8s)	9.76	0.4978	0.2667	0.6114	0.4788	0.5370
Faster R-CNN (ResNet50+FPN)	80.01	0.4223	-	0.5244	0.4831	0.5029

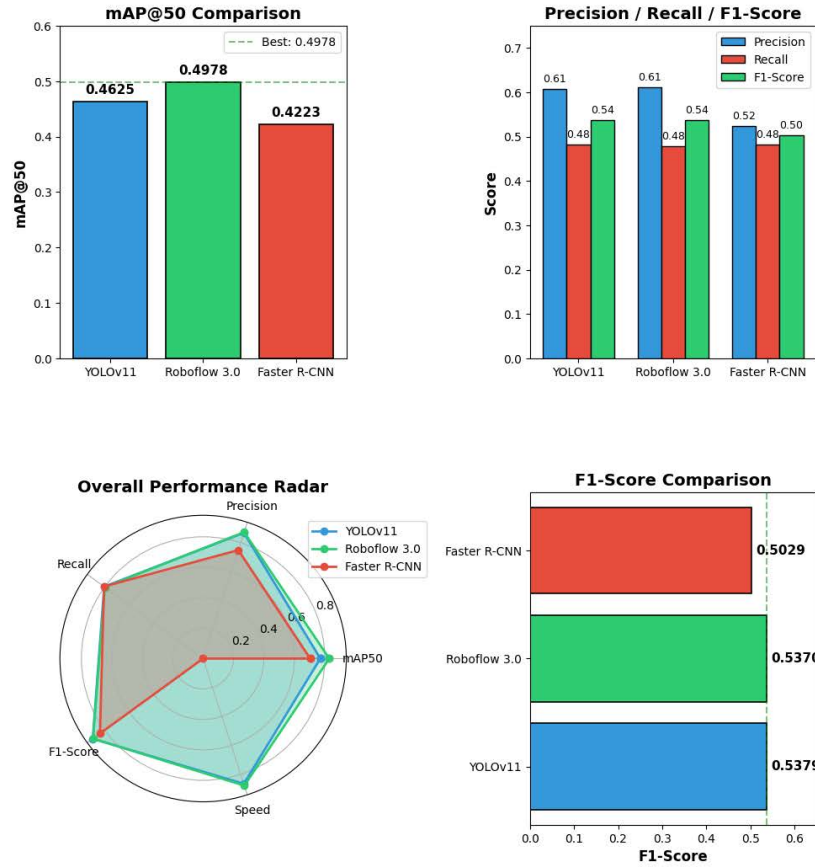


Figure 5.4: Weapon Detection Model Performance Comparison

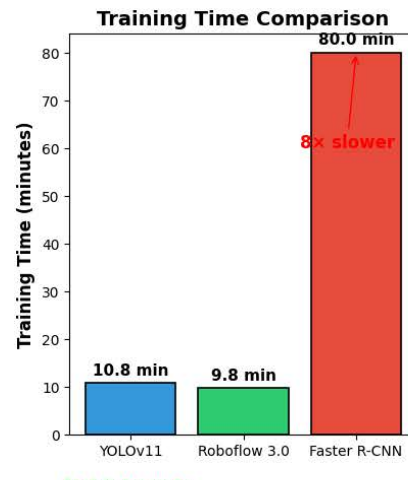


Figure 5.5: Training Time Comparison

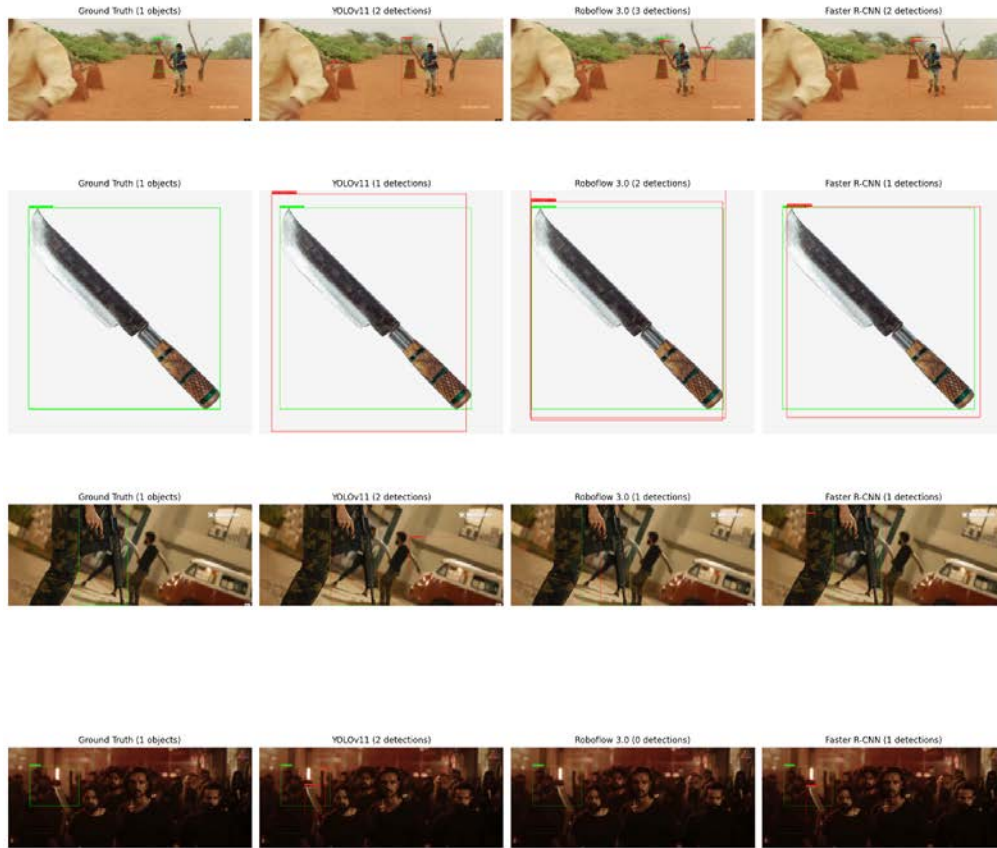


Figure 5.6: Ground Truth Vs. Model Predictions

5.3.4 Analysis

Key Findings:

1. **One-stage detectors outperform two-stage:** Both YOLO-based models achieved higher mAP@50 (0.46-0.50) compared to Faster R-CNN (0.42), while training 8× faster.

2. **Roboflow 3.0 achieves best overall performance:** Highest mAP@50 (0.4978), best precision (0.6114), and fastest training time (9.76 minutes), making it the recommended model for deployment.
3. **Speed-accuracy trade-off:** One-stage detectors provide the optimal balance for real-time CCTV applications, with inference times suitable for deployment on edge devices.
4. **Two-stage detector limitations:** Despite higher parameter count (41.3M vs 9-11M), Faster R-CNN showed lower accuracy and significantly longer training time (80 minutes vs 10 minutes), indicating diminishing returns for this dataset size.

Per-Class Performance Analysis:

Table 5.6: Weapon Detection Results on Indian Test Set

Class	mAP@50	Notes
Firearm	0.604	Best detection - distinct shape features
Blunt Weapon	0.499	Moderate - varied appearances (lathi, rod, bat)
Bladed Weapon	0.390	Most challenging - smaller objects, occlusion

5.3.5 Model Selection for Deployment

Based on the comparative analysis, **Roboflow 3.0 (YOLOv8s)** is selected for the final violence detection system deployment due to:

1. **Highest mAP@50** (0.4978) among all evaluated models
2. **Best precision** (0.6114), minimizing false alarms in surveillance applications
3. **Fastest training** (9.76 minutes), enabling efficient model updates

4. **Real-time inference capability** suitable for CCTV deployment
5. **Production-ready architecture** with extensive community support

6. Scene Graph Analysis

6.1 Purpose

Scene graph analysis extracts interpretable spatial and temporal relationships between detected entities. Unlike black-box deep learning classifiers, scene graphs provide explainable features for operator review and alert contextualization.

6.2 Graph Structure

Nodes: People (detected by YOLOv8n person detector), Weapons (from custom weapon detector)

Edges: Spatial proximity relationships between node pairs

Motion: Optical flow-based intensity per region

4. Scene Graph Analysis

Avg People/Frame	Avg Weapons/Frame	People Close	Graphs Analyzed
3.6	0.6	2.2	8

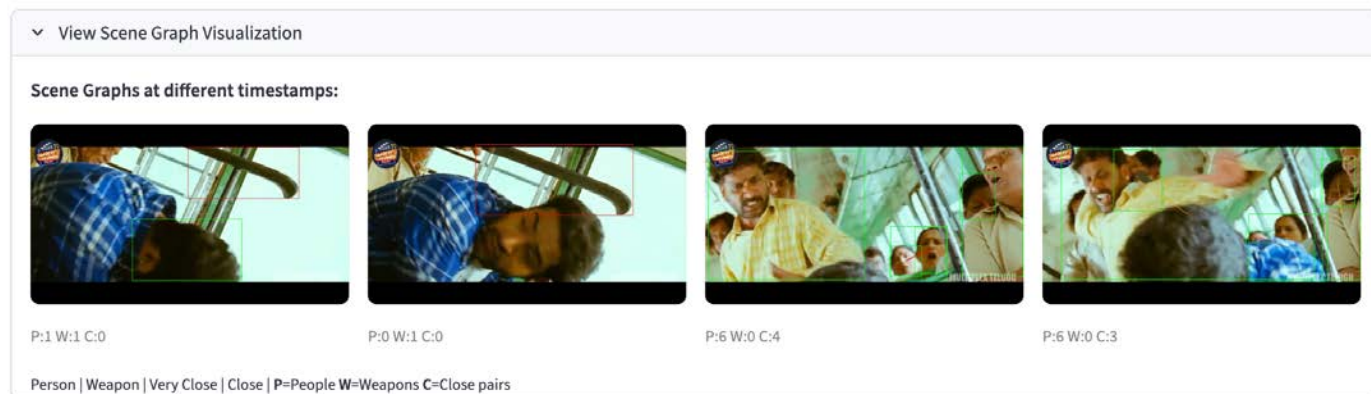


Figure 6.1: Example scene graph with proximity relationships and motion features

6.3 Feature Extraction

6.3.1 Proximity Features

- Normalized Euclidean distance between bounding box centers (0-1 scale).
- Person-person proximity for crowd analysis.
- Person-weapon proximity for threat assessment.

6.3.2 Motion Features

- Farneback optical flow computed between consecutive frames.
- Per-region motion magnitude within each bounding box.
- Motion direction (left/right/up/down/stationary).
- Approaching behavior detection based on relative motion.

6.4 Optical Flow Configuration

Dense optical flow is computed using the Farneback algorithm with parameters selected for real-time performance:

Table 6.1: Weapon Detection Results on Indian Test Set

Parameter	Value	Description	Rationale
Pyramid Scale	0.5	Image scale between pyramid levels	Standard multi-scale
Levels	3	Number of pyramid layers	Balance detail vs speed
Window Size	15	Averaging window size in pixels	Captures body-scale motion
Iterations	3	Refinement iterations per level	Convergence stability
Poly N	5	Pixel neighborhood for polynomial expansion	OpenCV default
Poly Sigma	1.2	Gaussian std for polynomial smoothing	OpenCV default
Gaussian Blur	5×5	Pre-processing kernel size	Noise reduction
Motion History	10	Frames retained for trend analysis	Escalation detection window

6.5 Threshold Calibration

Proximity thresholds were empirically calibrated using 100 videos (50 violent, 50 non-violent) from the Indian dataset:

Table 6.2: Calibrated Proximity Thresholds

Threshold	Value	Interpretation	Violent %	Non-violent %
VERY_CLOSE	0.15	Physical contact range	17.7%	17.8%
CLOSE	0.25	Striking range	37.5%	38.0%
NEAR	0.40	Threat proximity	64.2%	64.4%

Table 6.3: Calibrated Motion Thresholds

Threshold	Value	Violent %	Non-violent %
MODERATE_MOTION	2.0	73.0%	74.1%
HIGH_MOTION	5.0	53.9%	58.2%

6.6 Statistical Analysis

Mann-Whitney U tests revealed:

- Motion intensity: $p < 0.001$ (statistically significant discriminator)
- Person distance: $p = 0.795$ (high overlap, limited discriminative power)

Conclusion: Motion features discriminate better than proximity alone. Scene graphs provide explainability rather than primary classification power.

6.7 Rationale for Retaining Non-Discriminative Features

Despite limited standalone discriminative power, proximity features are retained for:

1. Explainability: Operators can understand 'two people very close with high motion'
2. Necessary condition: Close proximity is required (not sufficient) for physical violence
3. Multimodal synergy: Proximity + weapon detection = high threat
4. Alert context: Scene description enhances operator situational awareness

7. Action Classification

7.1 Purpose

The action classification module identifies specific violent actions to provide granular incident characterization beyond binary violence detection. The module employs an X3D-M video model pretrained on Kinetics-400 and adapts its outputs to a violence-specific action set.

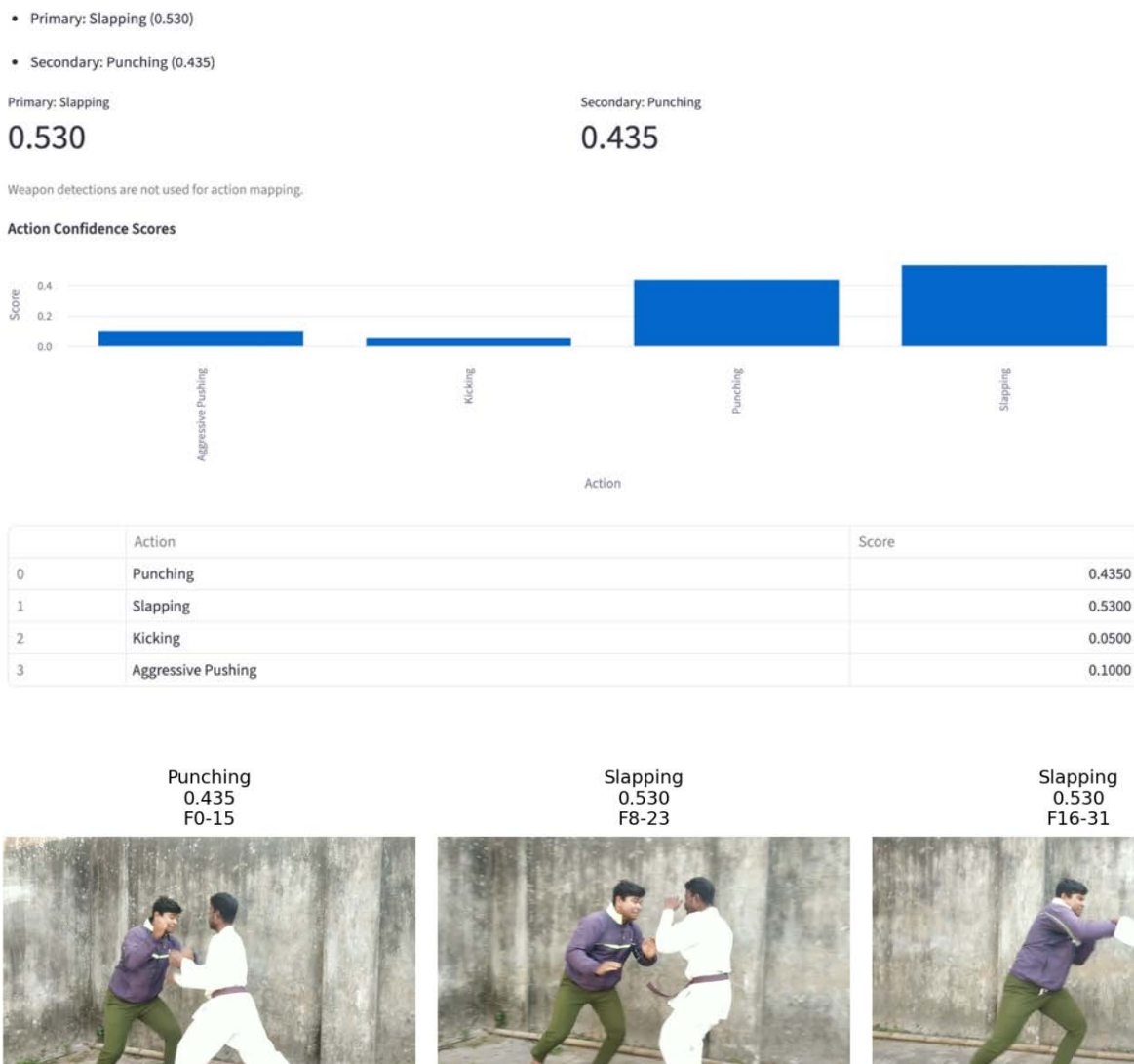


Figure 7.1: Action classification Output

7.1.1 X3D-M Architecture

X3D-M is used as a pretrained spatiotemporal video backbone for clip-level classification. In this implementation it operates on 16-frame clips at 256×256 resolution and outputs Kinetics-400 class probabilities.

7.1.2 Kinetics-400 Class Mapping

Kinetics-400 comprises 400 human action classes spanning sports, daily activities, and interactions. Since these classes do not directly correspond to violence-specific actions, a mapping layer translates relevant Kinetics predictions to the target action space:

Table 7.1: Kinetics-400 to Action Class Mapping

Kinetics Class ID	Kinetics Label	Mapped Action
79, 264, 265	boxing_punching, punching_bag, punching_person	punching
310	slapping	slapping
185	kicking_person	kicking
396, 178	wrestling, headbutting	pushing
339, 116	sword_fighting, fencing	weapon_attack

The top-20 softmax probabilities are extracted, and scores for mapped classes are aggregated with a 2× multiplier to emphasize violence-relevant predictions over the remaining 390+ non-violent classes.

7.1.3 Preprocessing Pipeline

Input videos are preprocessed to match Kinetics-400 training statistics:

1. **Frame sampling:** 16 frames uniformly sampled from the input clip
2. **Spatial resize:** Bilinear interpolation to 256×256
3. **Color normalization:** RGB values scaled to $[0, 1]$, then normalized with $\mu = 0.45$, $\sigma = 0.225$ per channel
4. **Tensor format:** $[B, C, T, H, W] = [1, 3, 16, 256, 256]$

7.1.4 Motion-Augmented Classification

Dense optical flow (Farneback) is computed across the clip to detect characteristic motion patterns. These motion cues boost or adjust the X3D-derived scores, especially when motion is strong and X3D confidence is weak.

Table 7.2: Motion Pattern Classification

Action	Motion Pattern	Detection Rationale
Slapping	Horizontal dominance (H/V > 1.3)	Slaps follow a lateral sweeping arc across the frame
Kicking	Vertical dominance (V/H > 1.6) + burst	Kicks produce upward leg motion with sudden acceleration
Punching	Impact burst + low horizontal ratio	Punches are forward thrusts with rapid deceleration on contact
Pushing	Sustained motion, no directional dominance	Grappling and shoving produce diffuse, prolonged motion

7.1.5 Configuration Parameters

Table 7.3: Motion Analysis Thresholds

Parameter	Value	Rationale
HORIZONTAL_DOMINANCE_RATIO	1.3	Empirically separates lateral sweeps from forward motion; lower values caused punch/slap confusion
VERTICAL_DOMINANCE_RATIO	1.6	Higher threshold prevents walking/running from triggering kick detection
IMPACT_MAGNITUDE_THRESHOLD	12.0	Optical flow magnitude indicating sudden deceleration consistent with strike impact
BURST_STD_RATIO	0.6	Motion standard deviation relative to mean; captures impulsive vs sustained motion
BURST_MAX_RATIO	2.0	Peak-to-average motion ratio indicating explosive movement
MIN_MOTION_THRESHOLD	1.0	Below this average flow magnitude, scene is effectively stationary

These thresholds were determined through iterative refinement on validation clips, balancing sensitivity to genuine violent actions against false positives from benign high-motion activities.

7.1.6 Weapon Integration

When the weapon detection module identifies a weapon in the scene, action classification is adjusted:

- weapon_attack score receives +0.4 boost
- punching and slapping scores are halved

This prevents misclassification of armed attacks as unarmed strikes and ensures appropriate threat characterization when weapons are present.

8. Results Summary

8.1 Violence Classification

Table 8.1: Violence Classification Summary

Metric	RLVS (Best)	Indian (Best)
Model	MobileNetV3-BiLSTM-Attention	MobileNetV3-BiLSTM-Attention
Test Accuracy	97.73%	80.43%
F1-Score	97.73%	80.20%
AUC-ROC	99.91%	85.63%
Inference Time	23.08 ms	27.78 ms
Parameters	7,241,011	7,241,011

Improvement over prior state-of-the-art: +6.73% vs Paper 1 (91%), +6.70% vs Paper 2 (91.03%)

8.2 Weapon Detection

Table 8.2: Weapon Detection Summary

Model	Type	Training Time (min)	mAP@50	Precision	Recall	F1-Score
YOLOv11 (Small)	One-stage	10.76	0.4625	0.6079	0.4824	0.5379
Roboflow 3.0 (YOLOv8s)	One-stage	9.76	0.4978	0.6114	0.4788	0.5370
Faster R-CNN (ResNet50 + FPN)	Two-stage	80.01	0.4223	0.5244	0.4831	0.5029

- Roboflow 3.0 achieves best mAP@50 (0.4978) with fastest training (9.76 min)
- One-stage detectors outperform two-stage by 18% mAP@50 while training 8× faster

8.3 Scene Graph Analysis

Table 8.3: Weapon Detection Summary

Metric	Value
Person Detector	YOLOv8n (COCO pretrained)
Optical Flow	Farneback (15×15 window)
Inference Time	~50 ms
Features Extracted	proximity, motion, approaching, weapon proximity
Statistical Discrimination	$p > 0.05$ (not independently discriminative)
Role	Explainability + contextual risk scoring

Note: Empirical analysis confirmed scene graph features serve as supplementary context for the rule-based risk module rather than standalone classifiers.

8.4 Action Classification

Table 8.4: Weapon Detection Summary

Metric	Value
Model	X3D-M (Kinetics-400 pretrained)
Input	16 frames, 256×256
Inference Time	~80 ms
Action Classes	punching, slapping, kicking, pushing, weapon_attack, no_action

Note: Quantitative accuracy not evaluated due to lack of action-labeled ground truth in Indian dataset. Module provides supplementary incident characterization rather than primary detection.

8.5 System Performance

Table 8.5: End-to-End System Performance

Component	Inference Time (32-frame window)
Violence Classifier	832.4 ms
Weapon Detector	3,434.3 ms
Action Classifier	1,432.3 ms
Scene Graph	7,413.3 ms
Total Pipeline	13,112.3 ms

The inference times in Table 8.5 were measured on CPU-based Streamlit deployment. The violence classifier achieves 832.4 ms per 32-frame window, enabling real-time detection at 1.28x video speed. Combined violence and weapon detection requires 4.27 seconds on CPU, which is not suitable for real-time operation. GPU benchmarks on NVIDIA Tesla T4 demonstrate 27.78 ms and 24.5 ms per inference for violence and weapon detection respectively, yielding ~52 ms combined latency capable of real-time performance. The MobileNetV3-Large backbone was selected for its edge compatibility, and the architecture supports deployment on CCTV infrastructure through ONNX conversion and TensorRT optimization on platforms such as NVIDIA Jetson. Action classifier and scene graph modules are designed for post-detection forensic analysis rather than continuous monitoring.

9. Conclusions and Recommendations

9.1 Summary of Contributions

This dissertation makes the following contributions to violence detection for Indian surveillance contexts:

1. State-of-the-art violence classification achieving 97.73% accuracy on RLVS benchmark, surpassing prior work by 6.7 percentage points.
2. Systematic quantification of domain gap for violence detection: 17.3% accuracy drop from RLVS to Indian context.
3. Creation of 676-image Indian weapon dataset with semi-automatic annotation using Grounding DINO.
4. Comprehensive comparison of three detection architectures (YOLOv11, Roboflow 3.0, Faster R-CNN) showing one-stage detectors achieve 18% higher mAP@50 than two-stage while training $8\times$ faster, with $6.7\times$ improvement over zero-shot baselines.
5. Empirically calibrated scene graph thresholds from 100-video statistical analysis.
6. Hybrid action classification combining X3D-M with motion pattern analysis.

9.2 Limitations

Several limitations constrain the current work:

- **Dataset size:** 676 weapon images and 291 violence videos limit model generalization.

- **Environmental diversity:** Limited nighttime and adverse weather footage
- **Single-camera:** No multi-camera tracking or cross-view fusion
- **Action classification:** No ground-truth action labels for quantitative evaluation; relies on Kinetics-400 transfer which may not cover India-specific violence patterns.
- **Weapon false positives:** Arms/hands in fighting poses occasionally misclassified as weapons.
- **Non-discriminative features:** Proximity thresholds showed near-identical distributions across violence/non-violence classes, limiting standalone predictive value.

9.3 Future Work

Recommended directions for future research:

1. Dataset expansion through active learning on unlabeled surveillance footage
2. Multi-camera tracking for comprehensive area coverage
3. Edge deployment optimization for NVIDIA Jetson or Intel NCS
4. Attention visualization for model explainability
5. Temporal action localization for incident timeline reconstruction
6. Integration with existing VMS (Video Management Systems)
7. Audio-visual fusion incorporating scream/shout detection for improved recall in occluded scenarios.

8. Graph neural networks for learned scene graph reasoning, potentially discovering discriminative relationship patterns beyond hand-crafted proximity thresholds.

9.4 Final Remarks

This work demonstrates that domain-specific training data is essential for practical violence detection deployment. No amount of model sophistication compensates for training-test distribution mismatch. Future surveillance AI systems must prioritize representative dataset collection capturing the diversity of real-world deployment environments.

Appendices

APPENDIX A

Implementation Details and Experimental Settings

This appendix provides supplementary implementation details, configurations, and supporting information that are not included in the main chapters to preserve readability, but are essential for reproducibility and deeper technical understanding of the proposed system.

A.1 Hardware and Software Environment

All experiments were conducted on a cloud-based GPU environment with the following specifications:

- **GPU:** NVIDIA Tesla T4 (15.8 GB VRAM)
- **CPU:** Intel Xeon (2.3 GHz)
- **RAM:** 32 GB
- **Operating System:** Ubuntu 20.04 LTS

Software Stack:

- Python 3.10
- PyTorch 2.x
- OpenCV 4.8
- Ultralytics YOLO Framework

- Roboflow SDK
- NumPy, SciPy, Matplotlib
- CUDA 11.8, cuDNN 8.x

A.2 Hyperparameter Summary

Violence Classification Models

- Optimizer: Adam
- Learning Rate:
 - Pretrained models: $1e-4$
 - 3D CNN (from scratch): $5e-4$
- Batch Size: 4
- Epochs: 30 (Early stopping with patience = 7)
- Loss Function: Cross-Entropy Loss

Weapon Detection Models

- Input Resolution: 640×640
- Optimizer: MuSGD (YOLO-based models), SGD (Faster R-CNN)
- Initial Learning Rate: 0.00143 (YOLO), 0.005 (Faster R-CNN)
- Epochs: 50 (YOLO), 30 (Faster R-CNN)

A.3 Dataset Annotation Guidelines

Weapon annotations were generated using a semi-automatic pipeline based on **Grounding DINO** with text prompts covering multiple Indian weapon variations (e.g., *lathi*, *sickle*, *machete*, *iron*

rod).

All bounding boxes were manually verified to ensure:

- Accurate object coverage
- Minimal background overlap
- Consistent labeling across frames

Annotations were exported in **YOLO format** for training compatibility.

A.4 Evaluation Metrics

The following evaluation metrics were used consistently across experiments:

- **Violence Classification:** Accuracy, Precision, Recall, F1-Score, AUC-ROC
- **Weapon Detection:** mAP@50, Precision, Recall, F1-Score
- **System Performance:** Inference latency (ms), FPS

These metrics were chosen to balance **detection accuracy** with **real-time feasibility** for surveillance deployment.

A.5 Reproducibility Notes

- All random seeds were fixed during training to ensure reproducibility.
- Model checkpoints were selected based on best validation performance.
- Dataset splits were mutually exclusive across train, validation, and test sets.
- No test data was used during hyperparameter tuning.

A.6 Ethical Considerations

The datasets used were sourced from publicly available videos and surveillance-style footage. No personally identifiable information was intentionally collected or released. The system is designed as a **decision-support tool** and not as a replacement for human judgment in security-critical scenarios.

References

- [1] K. Merit, M. Beladgham, and A. Taleb-Ahmed, "Temporal fusion strategy for violence detection: utilising convolutional and LSTM neural networks for surveillance videos," *Acta Polytechnica*, vol. 65, no. 3, pp. 306-319, 2025.
- [2] S. Senthilkumar, G. Agarwal, A. Shirish, and S. Kolte, "Real Time Violence Detection System Using YOLOv7 and Deep Learning Techniques," in *Proc. 3rd International Conference on Intelligent Data Communication Technologies and Internet of Things (IDCIoT)*, IEEE, 2025, pp. 1447-1454.
- [3] R. Jeevan and B. Avanthika, "Intelligent Video Surveillance Systems With Violence Detection," in *Proc. International Conference on Data Science, Agents & Artificial Intelligence (ICDSAAI)*, IEEE, 2025, pp. 1-6.
- [4] P. Sethi et al., "YOLOv7-based Violence Detection in Surveillance Videos," in *Proc. International Conference on Advances in Systems, Signals and Image Computing (ASSIC)*, IEEE, 2025.
- [5] S. Chakraborty, S. Zahir, N. T. Orchi, M. F. B. Hafiz, A. O. M. Shamsuddoha, and S. M. Dipto, "Violence detection: A multi-model approach towards automated video surveillance and public safety," in *Proc. International Conference on Advances in Computing, Communication, Electrical, and Smart Systems (iCACCESS)*, IEEE, 2024, pp. 1-6.
- [6] E. B. Varghese, A. Elzein, Y. Yang, and M. Qaraqe, "A temporal-spatial deep learning framework leveraging dynamic 3D attention maps for violence detection," *Neural Computing and Applications*, Springer, 2025, pp. 1-21.

- [7] J. H. Park, M. Mahmoud, and H. S. Kang, "Conv3D-based video violence detection network using optical flow and RGB data," *Sensors*, vol. 24, no. 2, p. 317, 2024.
- [8] Ultralytics, "YOLOv8: State-of-the-art object detection," 2024. [Online]. Available: <https://github.com/ultralytics/ultralytics>
- [9] S. Liu et al., "Grounding DINO: Marrying DINO with Grounded Pre-Training for Open-Set Object Detection," *arXiv preprint arXiv:2303.05499*, 2023.

LIST OF SYMBOLS AND ABBREVIATIONS

Abbreviation	Full Form
AUC	Area Under Curve
BiLSTM	Bidirectional Long Short-Term Memory
C3D	Convolutional 3D
CCTV	Closed-Circuit Television
CNN	Convolutional Neural Network
COCO	Common Objects in Context
CPU	Central Processing Unit
DNN	Deep Neural Network
FN	False Negative
FP	False Positive
FPN	Feature Pyramid Network
FPS	Frames Per Second
GPU	Graphics Processing Unit
LSTM	Long Short-Term Memory
mAP	mean Average Precision
RLVS	Real Life Violence Situations
ROC	Receiver Operating Characteristic
ROI	Region of Interest
RT-DETR	Real-Time Detection Transformer
SSD	Single Shot Detector
TN	True Negative
TP	True Positive
X3D	Expand 3D Network
YOLO	You Only Look Once
μ	Mean
σ	Standard Deviation

Glossary

AUC-ROC: Area Under the Receiver Operating Characteristic Curve - measures classification performance across all thresholds.

BiLSTM: Bidirectional Long Short-Term Memory - recurrent neural network processing sequences in both directions.

Burst motion: Short, high-magnitude motion indicating sudden impacts.

COCO: Common Objects in Context - standard object detection benchmark dataset.

CNN: Convolutional Neural Network - deep learning architecture for image processing.

F1 Score: Harmonic mean of precision and recall.

FPS: Frames Per Second - video processing throughput.

IoU: Intersection over Union - overlap metric for bounding boxes.

Kinetics-400: A dataset of 400 human action classes used to pretrain video models.

LSTM: Long Short-Term Memory - recurrent neural network for sequential data.

mAP: mean Average Precision - standard object detection metric.

Optical Flow: Motion estimation between consecutive video frames.

RLVS: Real-Life Violence Situations dataset.

X3D-M: A pretrained spatiotemporal video model used to classify actions in short clips.

YOLO: You Only Look Once - real-time object detection architecture.

Checklist of Items for the Final Dissertation / Project / Project Work Report

•	Is the final report neatly formatted with all the elements required for a technical Report?	Yes
•	Is the Cover page in proper format as given in Annexure A?	Yes
•	Is the Title page (Inner cover page) in proper format?	Yes
•	(a) Is the Certificate from the Supervisor in proper format?	Yes
	(b) Has it been signed by the Supervisor?	Yes
•	Is the Abstract included in the report properly written within one page? Have the technical keywords been specified properly?	Yes Yes
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•	Does the Table of Contents include page numbers?	Yes
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	Are all the references cited in the body of the report	Yes
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I certify that I have properly verified all the items in this checklist and ensure that the report is in proper format as specified in the course handout.

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Date: **01/02/2026**

A. Kaustiki

Signature of the Student

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